

Graph Neural Network Features for Dynamic S&P 500 Portfolio Construction

WRDS/CRSP Dynamic Constituents, Monthly Rebalancing, 2015–2024 OOS

Project Group 6

Course Project Report

Executive Summary

This project evaluates whether graph neural networks (GNNs) can improve portfolio construction on a dynamic S&P 500 universe. The data are built from WRDS/CRSP daily stock data and historical S&P 500 membership [2, 10]. Each stock is represented by a CRSP PERMNO, and at each monthly rebalance date the active index constituents form the investable universe. A rolling correlation graph is rebuilt every month, so the graph reflects changing market relationships rather than a static industry taxonomy.

The research design has three stages. **Stage 1** tests GCN, GraphSAGE, GAT, and APPNP as direct stock-ranking portfolio models. To address the concern that the original six features may be too sparse, Stage 1 also includes a 10-feature robustness check. **Stage 2** changes the role of GNNs: the GCN is used as a graph-aware feature extractor, and its score and hidden embeddings are fed into a Random Forest (RF) rank model. **Stage 3** builds the final frozen strategy: RF + GCN score/embeddings, a five-seed RF ensemble, long-only equal-weight top 20 selection, and 10 bps transaction costs.

Table 1: Headline OOS performance, 119 monthly rebalances

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover
RF + GCN Final	12.33%	18.24%	0.676	-24.99%	1.60
RF Base	10.44%	18.28%	0.571	-29.25%	1.71
EqualWeight	10.66%	18.06%	0.590	-31.48%	0.02
Momentum60	5.50%	20.14%	0.273	-34.10%	1.29

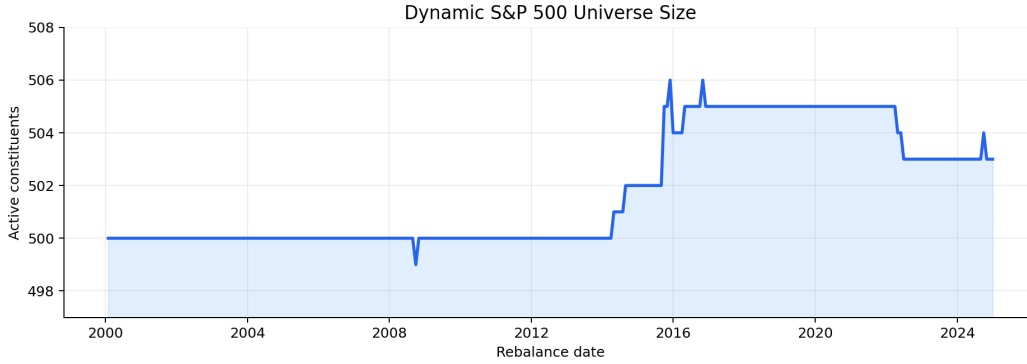
The main conclusion is not that pure GNNs are universally strong asset allocators. In fact, direct GNN portfolios are mixed and do not consistently dominate EqualWeight. The stronger finding is that GCN representations add useful relational information to a conventional RF model. The final frozen RF + GCN strategy improves Sharpe and maximum drawdown relative to RF Base, EqualWeight, and Momentum60.

1 Data Construction

The data layer uses local WRDS/CRSP daily stock files and a historical S&P 500 membership file [2, 10]. PERMNOs are used instead of tickers to avoid ticker changes and ticker reuse. The universe is dynamic: for each monthly rebalance date, the strategy only considers stocks that are active S&P 500 constituents at that time. This design mitigates constituent survivorship bias relative to selecting today’s index members and backfilling them through history.

Table 2: WRDS/CRSP data audit

Item	Value
CRSP daily date range	1999-07-06 to 2024-12-31
Project sample	2000-01-01 to 2024-12-31
OOS period	2015-01-30 to 2024-11-29
CRSP daily rows	4,788,231
Distinct PERMNOs in daily data	1,110
Monthly active S&P 500 constituents	499–506
OOS graph snapshots	119

**Figure 1:** Historical active S&P 500 constituent count by month. The universe remains close to 500 stocks while allowing entries, deletions, and share-class mechanics.

The data are not claimed to be production-trading ready. Transaction costs are simplified, sector information is not used, and securities with insufficient rolling lookback data are excluded from the corresponding graph snapshots. However, for a course project, using historical index membership and CRSP identifiers is a meaningful improvement over a fixed survivor-only ticker list.

2 Features, Labels, and Dynamic Graphs

The baseline node feature vector contains six interpretable price-volume variables:

$$x_{i,t} = [\text{Mom}_{20}, \text{Mom}_{60}, \text{Vol}_{20}, \text{Vol}_{60}, \Delta\text{Volume}_{20}, \beta_{\text{MKT},60}].$$

Momentum is calculated from adjusted prices, volatility is the rolling standard deviation of daily returns, volume change captures short-run trading activity, and beta is estimated against an equal-weight market proxy. The beta proxy is computed in a leave-one-out style so that a stock’s own return does not mechanically enter its market benchmark.

The prediction target is the next-20-trading-day forward return with a one-day execution lag:

$$y_{i,t} = \frac{P_{i,t+21}}{P_{i,t+1}} - 1.$$

The one-day lag avoids using the signal-date close as both the information set and the execution price. All features and graph edges use only information available by date t .

At each rebalance date, the graph is reconstructed from the most recent 60 daily returns. Each stock keeps its top $k = 10$ positively correlated neighbors:

$$A_{ij,t} = \begin{cases} \rho_{ij,t}^{(60)}, & j \in \text{TopK}_{10}^+(i, t), \\ 0, & \text{otherwise.} \end{cases}$$

This creates a sparse dynamic graph. Correlation values are used as edge weights for GCN and GAT-style weighted message passing, while the edge set itself defines which stocks exchange information. The motivation is consistent with relational stock-ranking research, where inter-firm links can carry information that is missed when each stock is modeled independently [3].

3 GNN Message-Passing Models

Let $H_t^{(0)} = X_t \in \mathbb{R}^{N_t \times F}$ be the node feature matrix at rebalance date t , and let $G_t = (V_t, E_t, A_t)$ be the dynamic stock graph. Each model outputs one score per stock, $\hat{y}_{i,t}$, which is used for cross-sectional ranking rather than as a calibrated return forecast. The four architectures follow the standard GNN families introduced in GCN, GraphSAGE, GAT, and APPNP [4–6, 9].

GCN. With self-loops $\tilde{A} = A + I$ and degree matrix $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, one GCN layer is

$$H^{(\ell+1)} = \phi\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)}\right).$$

Financially, GCN smooths each stock's signal over its correlated neighbors. This is useful when individual stock labels are noisy but related stocks share sector, style, or macro exposures.

GraphSAGE. GraphSAGE separates a stock's own representation from an aggregated neighbor representation:

$$m_i^{(\ell)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j^{(\ell)}, \quad h_i^{(\ell+1)} = \phi(W_s h_i^{(\ell)} + W_n m_i^{(\ell)}).$$

Because the same aggregation rule is reused across changing node sets, GraphSAGE is naturally suited to a dynamic index universe.

GAT. GAT learns neighbor-specific attention weights:

$$e_{ij}^{(k)} = \text{LeakyReLU}\left(a_k^\top [W_k h_i \| W_k h_j]\right), \quad \alpha_{ij}^{(k)} = \frac{\exp(e_{ij}^{(k)})}{\sum_{r \in \mathcal{N}(i)} \exp(e_{ir}^{(k)})},$$

$$h_i^{(\ell+1)} = \parallel_{k=1}^K \phi\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} W_k h_j\right).$$

Attention can change by year and by rebalance date because both features and graph neighborhoods change through time.

APPNP. APPNP first computes an MLP seed $Z = f_\theta(X)$, then performs personalized PageRank-style propagation:

$$H^{(0)} = Z, \quad H^{(r+1)} = (1 - \alpha) \hat{A} H^{(r)} + \alpha Z.$$

The teleport term prevents excessive drift from the original stock-level signal, while repeated propagation diffuses information through the graph.

All direct GNNs are trained with a pairwise ranking loss:

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \log(1 + \exp[-s_{ij}(\hat{y}_{i,t} - \hat{y}_{j,t})]), \quad s_{ij} = \text{sign}(y_{i,t} - y_{j,t}).$$

Validation uses cross-sectional rank IC, the Spearman correlation between predicted scores and realized forward returns.

4 Walk-Forward Backtest Protocol

The backtest uses yearly walk-forward retraining. For test year Y , models train on historical snapshots through $Y - 2$, validate on $Y - 1$, and test on Y . This prevents using future validation information to choose a model for an earlier test year. Monthly returns are non-overlapping: the portfolio enters one trading day after the signal date and exits at the next rebalance entry date.

All model-based strategies rank active constituents each month. The standard portfolio rule is long-only top 20, equal-weight, with 10 bps transaction cost per unit of absolute turnover. EqualWeight holds all active graph nodes, while Momentum60 ranks stocks by past 60-day momentum and also holds the top 20. Performance is evaluated with standard risk-return measures such as volatility, maximum drawdown, and the Sharpe ratio [7, 8].

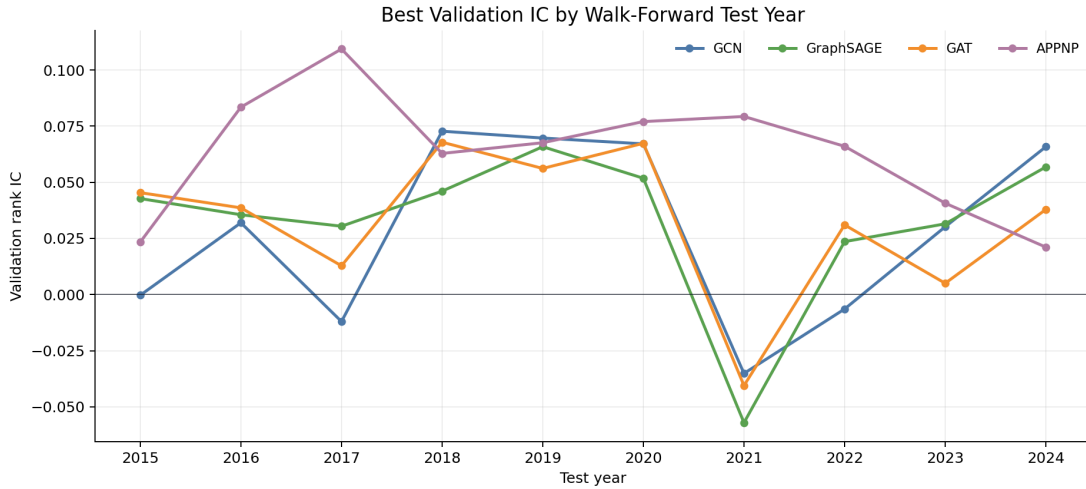


Figure 2: Best validation rank IC by walk-forward fold. IC values are modest, as expected for monthly stock-return prediction, but they diagnose whether each model learns a repeatable ranking signal.

5 Stage 1: Direct GNN Portfolio Models

The first experiment evaluates whether GNNs alone are good portfolio managers. The answer is mixed. With the original six features, GCN and GraphSAGE produce positive OOS performance, but neither clearly dominates EqualWeight. GAT is weaker, and APPNP suffers from a large drawdown.

Table 3: Stage 1: pure GNNs with original six features

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover
GCN	10.31%	18.99%	0.543	-26.54%	1.64
GraphSAGE	10.14%	17.16%	0.591	-29.56%	1.66
GAT	7.65%	18.59%	0.411	-28.62%	1.60
APPNP	4.95%	24.19%	0.204	-43.88%	1.61
EqualWeight	10.66%	18.06%	0.590	-31.48%	0.02
Momentum60	5.50%	20.14%	0.273	-34.10%	1.29

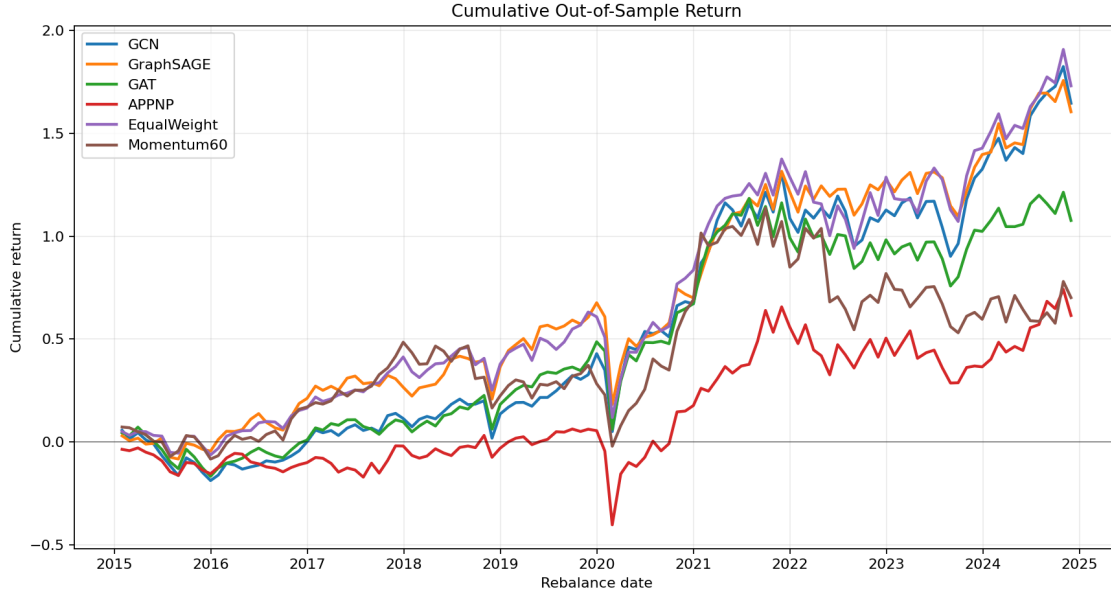


Figure 3: Cumulative OOS returns for direct GNN strategies and benchmarks using the six-feature specification.

To test whether pure GNN weakness is caused by too few inputs, I also run a 10-feature GNN robustness experiment. The expanded set adds medium-term and long-term momentum, downside volatility, idiosyncratic volatility, a longer beta, and volume surge. The result does not support a simple “more features solves it” conclusion.

Table 4: Stage 1 robustness: pure GNNs with 10 features

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Interpretation
GCN	3.76%	19.49%	0.193	-34.80%	Deteriorates sharply.
GraphSAGE	6.11%	19.37%	0.316	-35.60%	Deteriorates sharply.
GAT	10.28%	19.35%	0.531	-34.59%	Improves, but drawdown remains high.
APPNP	12.56%	29.22%	0.430	-52.85%	High return but unstable risk.
EqualWeight	10.68%	18.05%	0.592	-31.44%	Still difficult to beat.

The 10-feature results are useful because they strengthen the Stage 1 conclusion. Some architectures benefit from richer inputs, but improvement is not uniform and risk control remains weak. Therefore, pure GNNs are not adopted as the final portfolio construction method.

6 Stage 2: GCN as a Feature Extractor for RF

The second stage uses GCN differently. Rather than treating GCN as the final portfolio model, the GCN is trained on the dynamic graph and then used to generate a scalar score and a 64-dimensional hidden embedding. These outputs are concatenated with the original six stock features and passed into Random Forest, an ensemble tree method introduced by Breiman [1]:

$$z_{i,t} = \left[x_{i,t} \parallel \hat{y}_{i,t}^{\text{GCN}} \parallel h_{i,t}^{\text{GCN}} \right], \quad \hat{s}_{i,t} = f_{\text{RF}}(z_{i,t}).$$

The RF target is the within-month rank transform of realized forward returns,

$$q_{i,t} = \text{rank}_{i \in V_t}(y_{i,t})/N_t - 0.5.$$

This target matches the portfolio objective because the strategy only needs to rank stocks, not forecast returns in percentage points.

Table 5: Stage 2: RF baseline versus RF with GCN representation features

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover
RF Base	8.97%	18.87%	0.475	-29.79%	1.69
RF + GCN Features	10.07%	18.10%	0.557	-26.32%	1.61
GCN Direct	10.31%	18.99%	0.543	-26.54%	1.64
EqualWeight	10.66%	18.06%	0.590	-31.48%	0.02
Momentum60	5.50%	20.14%	0.273	-34.10%	1.29

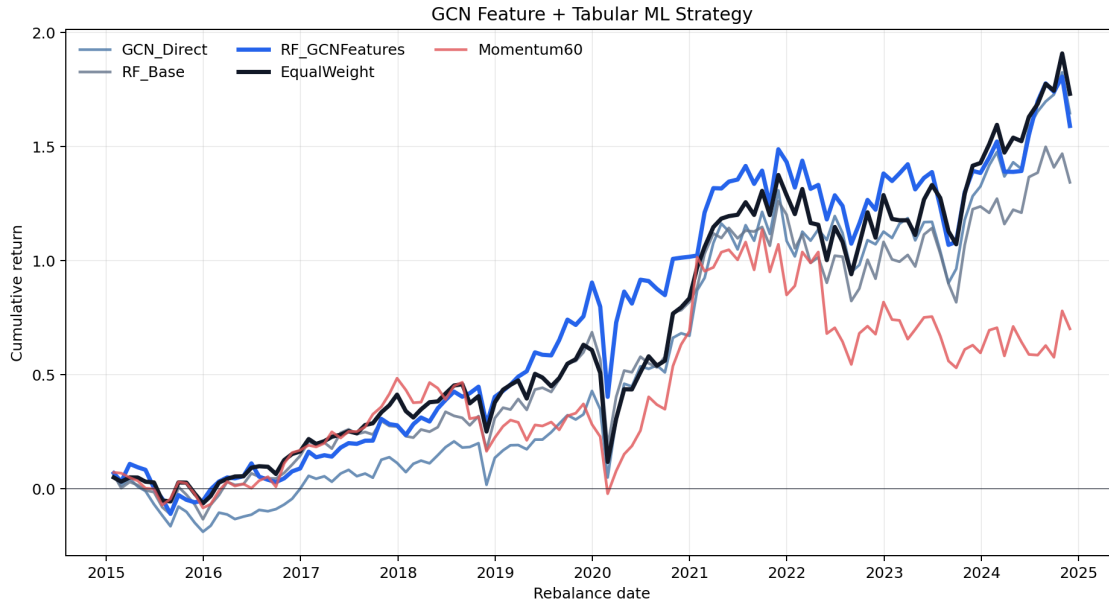


Figure 4: Stage 2 cumulative returns. RF + GCN Features improves over RF Base, showing that graph-derived representations contain incremental information for a tabular rank model.

The 10-feature hybrid check supports the same conclusion. In the expanded-feature setting, standalone RF has Sharpe 0.370, while RF + GCN Features reaches Sharpe 0.606. This suggests that the extra features are noisy when used independently by RF, but GCN can partially denoise and contextualize them through the correlation graph. Nevertheless, the 10-feature final strategy did not beat the six-feature final strategy, so the final model keeps the original six features for robustness and interpretability.

Table 6: Stage 2 feature-count robustness

Experiment	Strategy	Ann. Return	Sharpe	Max DD
6F	RF Base	8.97%	0.475	-29.79%
6F	RF + GCN Features	10.07%	0.557	-26.32%
10F	RF Base	7.16%	0.370	-36.09%
10F	RF + GCN Features	10.89%	0.606	-30.49%

7 Stage 3: Frozen RF + GCN Final Strategy

The final strategy is deliberately frozen before OOS evaluation. It does not perform annual hyperparameter selection, monthly rule switching, or long-short construction. Its pipeline is:

6F + GCN score + GCN embeddings $\rightarrow$ RF seed ensemble $\rightarrow$ long top 20, equal weight.

The five-seed RF ensemble trains the same RF specification with five random seeds and averages predicted scores:

$$\hat{s}_{i,t}^{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \hat{s}_{i,t}^{(m)}, \quad M = 5.$$

This is a variance-reduction device, not a new model family. RF randomness comes from bootstrap samples and random feature subsets; averaging across seeds makes the stock ranking less sensitive to one random draw.

Table 7: Final frozen strategy variants and benchmarks

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover
RF + GCN Final	12.33%	18.24%	0.676	-24.99%	1.60
RF + GCN EqualTop20	12.35%	18.47%	0.668	-26.82%	1.61
RF + GCN BufferTop20	11.97%	18.23%	0.657	-25.78%	1.52
RF + GCN VolTop20	12.02%	18.07%	0.665	-26.59%	1.64
RF Base	10.44%	18.28%	0.571	-29.25%	1.71
EqualWeight	10.66%	18.06%	0.590	-31.48%	0.02
Momentum60	5.50%	20.14%	0.273	-34.10%	1.29

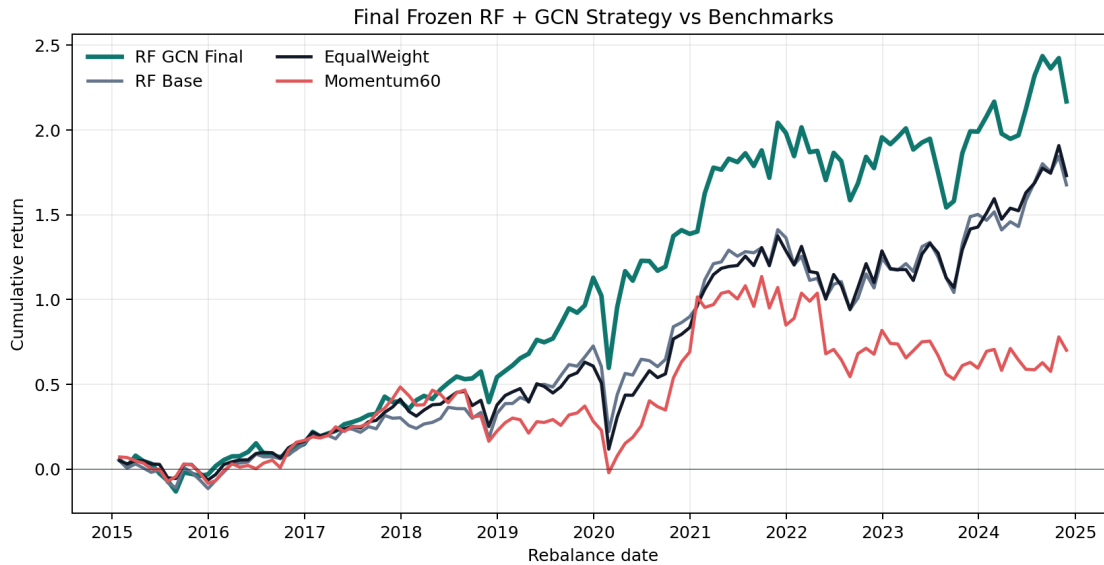


Figure 5: Final frozen RF + GCN strategy versus RF Base, EqualWeight, and Momentum60.

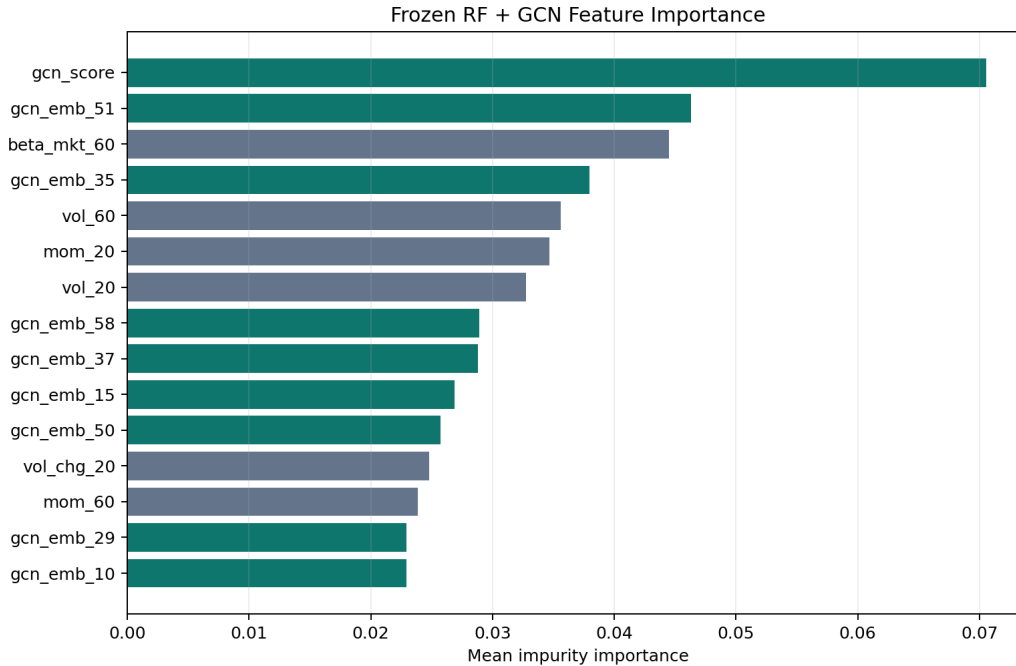


Figure 6: Final RF + GCN feature importance. Several GCN-derived dimensions rank alongside traditional stock features, supporting the interpretation that graph representations add information rather than merely changing the backtest rule.

The selected final version is the seed-ensemble RF + GCN strategy. Relative to the basic equal top-20 RF + GCN portfolio, it keeps annualized return almost unchanged, raises Sharpe from 0.668 to 0.676, and improves maximum drawdown from -26.82% to -24.99%. The holding-buffer and volatility-scaled variants are useful robustness checks, but neither improves the overall return-risk tradeoff enough to replace the seed ensemble.

8 Additional Robustness Checks

RF-specific feature expansion. After choosing the RF + GCN framework, I tested whether the final RF should receive a richer tabular feature set while keeping the GCN input fixed at the original six features. This is an important distinction: Stage 2 remains a clean ablation of whether GCN features help RF, while this test only asks whether the final production-style RF layer benefits from more stock-level predictors.

The refined RF feature set includes four momentum signals (21, 63, 126, and 252–21), five-day reversal, short- and medium-term volatility, downside volatility, 126-day beta, idiosyncratic volatility, volume surge, and dollar volume. All of these are computed from the existing daily price, return, and volume data, so no additional data download is required.

Table 8: Final-stage refined RF feature test

Strategy	Ann. Return	Sharpe	Max DD	Turnover
RF + GCN Final	12.33%	0.676	-24.99%	1.60
RF + GCN Refined Features	9.56%	0.512	-27.10%	1.57
RF Base	10.44%	0.571	-29.25%	1.71
RF Refined Base	6.96%	0.354	-36.22%	1.54
EqualWeight	10.66%	0.590	-31.48%	0.02

The refined features do not improve the final strategy. The standalone refined RF is also weaker than the original RF Base, suggesting that the additional predictors introduce noise or redundant splits in this sample. Therefore, the final strategy keeps the simpler six-feature specification and relies on GCN embeddings for relational information rather than expanding the tabular feature set.

Transaction costs and turnover. All reported strategy returns are net of transaction costs. The cost model subtracts

$$\text{Cost}_t = 0.001 \times \sum_i |w_{i,t} - w_{i,t-}|,$$

which corresponds to the baseline 10 bps per unit of absolute turnover. The final RF + GCN strategy has average monthly turnover around 1.60, implying an average monthly cost near 0.16%, or roughly 1.9% annualized before compounding. To check robustness to trading-cost assumptions, Table 9 recomputes performance at 0, 10, and 20 bps.

Table 9: Transaction cost sensitivity

Cost	Strategy	Ann. Return	Sharpe	Max DD
0 bps	RF + GCN Final	14.49%	0.794	-24.66%
0 bps	RF Base	12.72%	0.696	-28.95%
0 bps	EqualWeight	10.69%	0.592	-31.48%
0 bps	Momentum60	7.14%	0.354	-31.70%
10 bps	RF + GCN Final	12.33%	0.676	-24.99%
10 bps	RF Base	10.44%	0.571	-29.25%
10 bps	EqualWeight	10.66%	0.590	-31.48%
10 bps	Momentum60	5.50%	0.273	-34.10%
20 bps	RF + GCN Final	10.22%	0.560	-25.32%
20 bps	RF Base	8.20%	0.449	-29.55%
20 bps	EqualWeight	10.64%	0.589	-31.48%
20 bps	Momentum60	3.89%	0.193	-36.42%

The sensitivity analysis shows that RF + GCN remains strongest at the baseline 10 bps setting, while a more conservative 20 bps assumption materially reduces the advantage of active top-20 strategies. This cost sensitivity is an important limitation of the high-turnover design.

9 Interpretation

The project supports a nuanced conclusion. The dynamic stock graph is economically meaningful because correlated stocks transmit information about sector, style, and market-regime exposures, echoing the motivation in relational stock prediction [3]. However, directly turning a GNN score into a portfolio is not always robust. Pure GNNs face noisy labels, high turnover, and architecture sensitivity. This is why Stage 1 does not choose a direct GNN as the final strategy.

The strongest result is representation learning. RF Base treats each stock mostly independently, while RF + GCN receives a graph-aware summary of how that stock sits inside the correlation network. The improvement from RF Base to RF + GCN Features indicates that GCN embeddings contain information not fully captured by momentum, volatility, volume, and beta alone. The final RF seed ensemble then stabilizes the RF ranking without changing the economic idea.

10 Limitations

- The OOS period contains 119 monthly rebalances, which is useful for a course project but still one historical sample.

- The graph is based on rolling correlations only; sector classifications, fundamentals, macro variables, and options information are not used.
- Transaction costs are included, but market impact, borrow constraints, and shorting costs are not modeled.
- EqualWeight is a difficult benchmark because it has broad market exposure and very low turnover; RF + GCN improves risk-adjusted performance but trades much more frequently.
- The analysis should be interpreted as academic evidence of graph-aware representation value, not as a production trading system.

11 Conclusion

This project builds a modular graph-based portfolio framework on historical dynamic S&P 500 membership. Stage 1 shows that pure GNNs are not consistently superior as direct portfolio construction models, even after a 10-feature expansion. Stage 2 shows that GCN outputs improve a Random Forest rank model, supporting the idea that GNNs are more useful here as relational feature extractors. Stage 3 turns this finding into a frozen final strategy: six original features, GCN score and embeddings, a five-seed RF ensemble, and long-only equal-weight top-20 monthly selection. Under the baseline 10 bps transaction-cost assumption, the final strategy achieves **12.33% annualized return, 0.676 Sharpe, and -24.99% maximum drawdown**, outperforming RF Base, EqualWeight, and Momentum60 on OOS risk-adjusted performance.

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